Project Design Phase

Problem – Solution Fit Template

Date	7 September 2026
Team ID	Individual
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Project Name	Digital Metro Ticket Booking System using ServiceNow
Maximum Marks	2 Marks

Problem – Solution Fit: The proposed ServiceNow solution addresses the metro ticketing problem by providing a standardized digital booking request, automated fare calculation, station mapping, and QR-code ticket generation.

1. CUSTOMER SEGMENT(S) CS

• Metro passengers / end users • Daily commuters • Occasional travelers • Station Managers • Metro Operations Team • IT Administrators

6. CUSTOMER CONSTRAINTS CC

• Waiting time at ticket counters • Dependence on manual ticketing • Fare calculation effort • Need for quick and accessible booking • Paper-ticket dependency • Limited visibility of digital booking status

5. AVAILABLE SOLUTION(S) AS

• Manual ticket counters • Paper-based metro tickets • Existing digital/mobile ticketing options • QR-based ticketing • Digital payment methods • ServiceNow-based booking workflow

2. JOBS-TO-BE-DONE / PROBLEMS J&P;

• Book a metro ticket quickly • Select source and destination • Specify passenger type and quantity • Calculate the correct fare • Receive a usable digital ticket • Access the QR code when needed

9. PROBLEM ROOT CAUSE RC

• Manual counter-based booking • Separate steps for fare calculation • Paper ticket generation • Lack of a unified digital workflow • Manual handling can increase errors • Limited centralized booking data

7. BEHAVIOUR BE

• Passengers wait in queues • Passengers seek faster booking options • Users prefer digital payment methods • Users want immediate ticket confirmation • QR tickets are easier to carry • Users may avoid complex booking steps

3. TRIGGERS TR

• Need to travel by metro • Daily or occasional commute • Need to book multiple tickets • Peak-hour travel • Need for quick ticket access • Preference for cashless booking

10. YOUR SOLUTION SL

• ServiceNow Service Catalog ticket booking • Variables: Source, Destination, Passenger Type, Number of Tickets • Automated fare calculation • Station details mapping • Digital ticket generation • QR-code ticket generation • Booking records for tracking and analysis

8. CHANNELS OF BEHAVIOUR CH

8.1 ONLINE • ServiceNow Service Portal • Digital ticket interface • QR-code ticket access • Digital payment channels 8.2 OFFLINE • Metro station entry/validation • Station support when required • QR-code scanning at station

Problem – Solution Fit

The problem is addressed by replacing manual ticket-counter dependency with a standardized ServiceNow digital booking process. The solution captures source, destination, passenger type, and ticket quantity, maps station details, calculates the fare automatically, generates a digital ticket with a QR code, and maintains booking information for tracking and operational analysis.